

Development of Deep Neural Network Based Methods to Estimate Biological Age from Electrocardiograms

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Introduction

- Chronological age is defined as the time passed since birth.
- Biological age is defined as the biophysiological decline in organismal function and is influenced by lifestyle, environmental factors, genetics, disease, etc.
- The electrocardiogram (ECG) is a low-cost and widely available exam for screening and evaluation of cardiovascular diseases.
- Recently, artificial intelligence models have been developed to diagnose disease from ECG waveforms. One such method uses ECGs to estimate biological age.

Study Aim: To assess the use of neural network estimated biological age to predict cardiovascular disease.

Methods

Neural Networks

- In its most basic form, a neural network is a set of interconnected nodes, often referred to as neurons.
- Deep neural networks (DNNs) build on neural networks by incorporating multilayered sets of neurons.
- Convolutional neural networks (CNNs) are a specialized type of DNN designed for analyzing data such as images or time series by examining specific regions of the data.

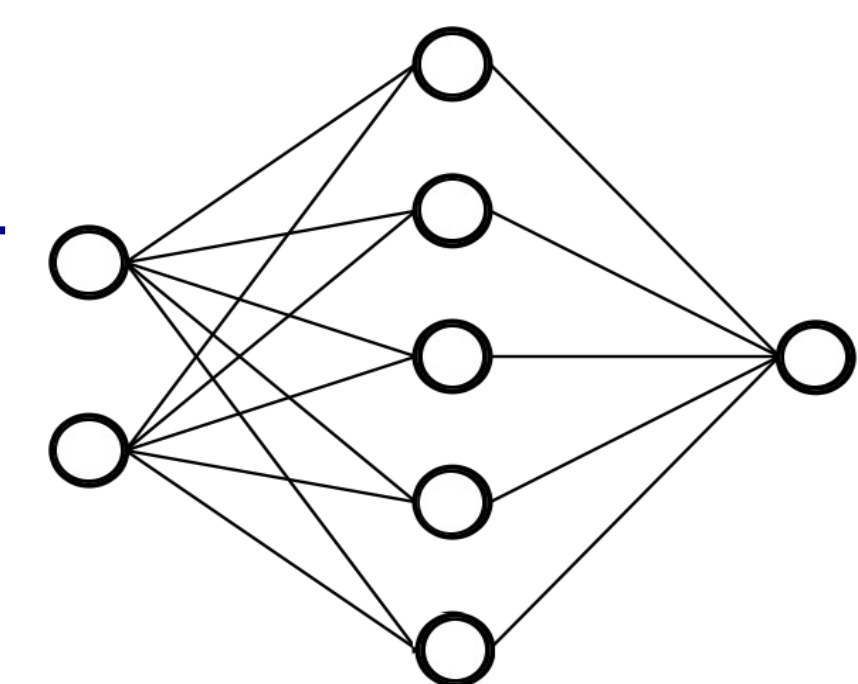


Figure A: A neural network

Datasets

- In total, we have access to over 400,000 ECGs from a variety of sources.
- UK BioBank: a large-scale database containing health information from half a million UK participants (54,919 ECGs).
- CODE 15: an ECG dataset of patients from Minas Gerais, Brazil (345,797 ECGs).

Methods (cont.)

Datasets (cont.)

- Sami-Trop: a cohort of patients with chronic Chagas cardiomyopathy (1,631 ECGs).
- PTB-XL: a large publicly available electrocardiography dataset (21,799 ECGs).

ECG-age

- Using a CNN framework applied to ECG waveforms, we aim to predict biological age.

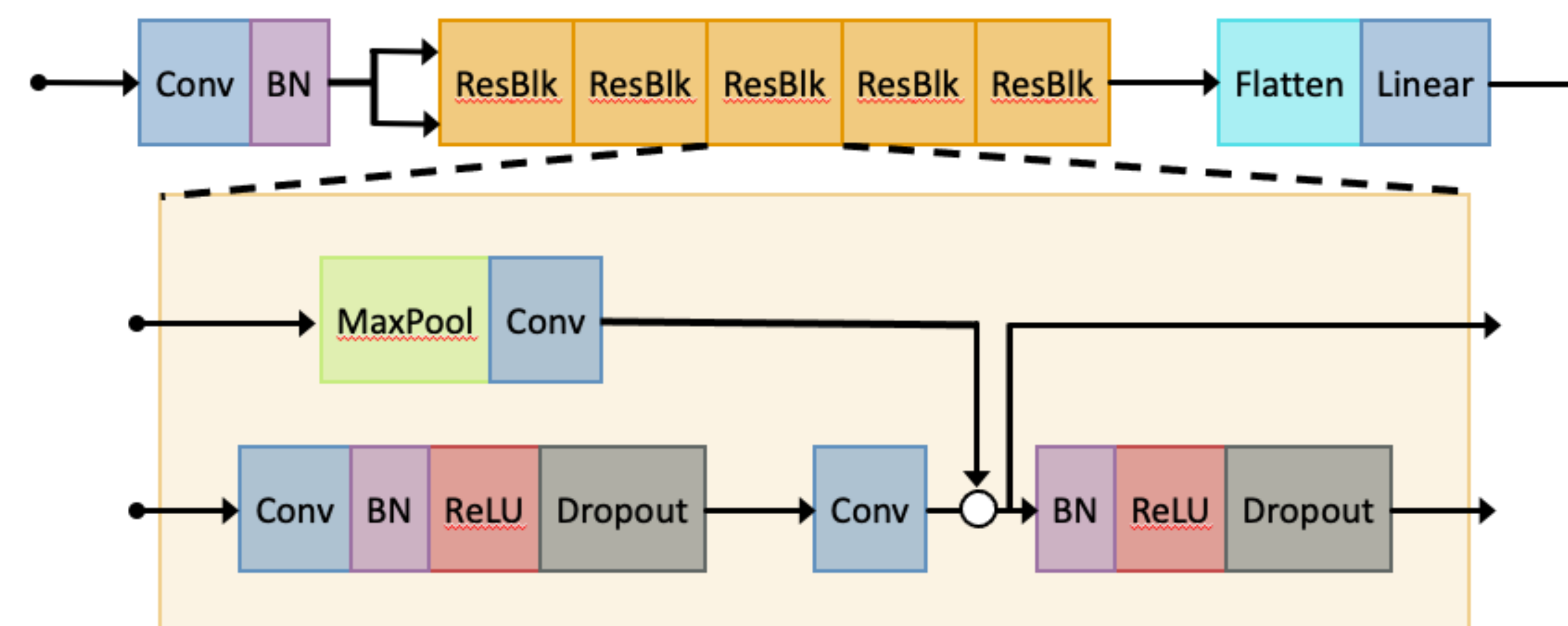


Figure B: ECG-age convolutional neural network framework

- The CNN is trained on a set of ECGs, and the estimated age is computed on a previously unseen subset of the ECGs from the UK BioBank.
- The performance of the model is determined by examining the Pearson correlation and the mean absolute error.

Results

- Several different CNNs were trained on the data varying the number of patients in the training and testing subsets.
- CNNs were trained using the UK BioBank dataset.
- Estimated ECG-age demonstrated a moderate correlation with chronological age of 0.51.
- The estimated ECG-age had a mean absolute error of 6.7.

Results (cont.)

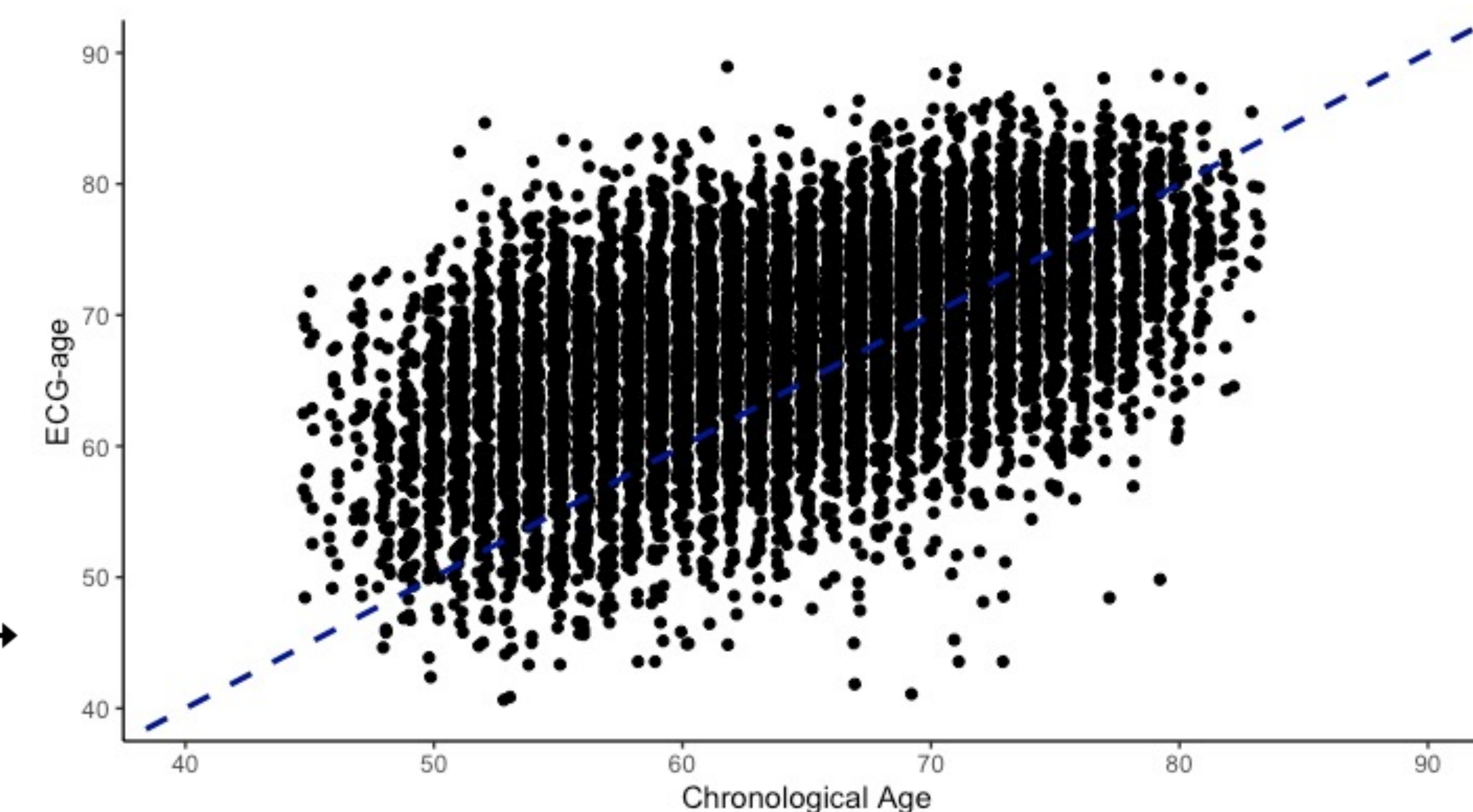


Figure C: Correlation between chronological age and deep neural network estimated electrocardiographic age

Next Steps

- Compute study population demographic information.
- Train the CNN using the entire available dataset. Currently, the model is trained using only the UK BioBank data.
- Evaluate the relationship between ECG-age and cardiovascular disease.

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